

Personalised Practice: Coordinate Geometry

Topic: Coordinate Geometry

Original examination-style practice material

Total: 12 marks

1 (a) $-3/5$ (b) (2, 2)

Gradient = $(-1 - 5) / (7 - (-3)) = -6/10 = -3/5$.

Midpoint = $((-3 + 7)/2, (5 + (-1))/2) = (2, 2)$.

Marks: correct signed differences · correct gradient · correct midpoint [3]

Accept: -0.6 for the gradient

2 $y = -2x + 3$

$m = (-5 - 7)/(4 - (-2)) = -12/6 = -2$.

Using P: $7 = -2(-2) + c$, so $c = 3$.

Marks: gradient method · $m = -2$ · substitution · equation [4]

3 $y = -1/2 x + 7$

Perpendicular gradient = $-1/2$.

$4 = -1/2(6) + c$, so $c = 7$.

Marks: negative reciprocal · uses point · finds intercept · forms equation · correct final form [5]

Accept: $x + 2y = 14$
